

Scala: The Powerhouse of Apache Spark

Scala, which is an acronym for Scalable Language, is a multi-paradigm, statically-typed, type-safe programming language focused on Web services. Widely used by data scientists today, its popularity is set to soar in the future because of the boom in the Big Data and data science domains.



The world is being flooded with data from a wide range of sources. The hottest trends in technology currently are Big Data and data science, both of which offer ways to cope with this data deluge. Many platforms have emerged in this space, but Apache Spark and Scala work in synergy to address the various challenges this humongous data throws up. They are being used on Facebook, Pinterest, NetFlix, Conviva and TripAdvisor, among others, for Big Data and machine learning applications.

So what is Scala?

Scala stands for Scalable Language. It was developed as an object-oriented and functional programming language. Everything in Scala is an object, even its primitive data types. If you write a snippet of code in Scala, you will see that the style is similar to a scripting language. It is very powerful, yet compact, and requires only a fraction of the lines of code compared to other commercially used languages. Due to its characteristics and support

for distributed/concurrent programming, it is popularly used for data streaming, batch processing, AWS Lambda Expression and analysis in Apache Spark. It is one of the most widely used languages by data scientists, and its popularity will soar in the future due to the boom in the Big Data and data science domains.

What is Apache Spark?

Apache Spark is a cluster computing framework based on Hadoop's MapReduce framework. Spark has in-memory cluster computing, which helps speed up computation by reducing the IO transfer time. It is widely used to deal with Big Data problems because of its distributed architectural support and parallel processing capabilities. It is preferred to Hadoop due to its stream processing and interactive query features. To provide a wide range of services, it has built-in libraries like GraphX, SparkSQL and MLlib. Spark supports Python, Scala, Java and R as programming languages, of which Scala is the most preferred.